



Deep reinforcement learning-based adaptive collision avoidance method for UAV in joint operational airspace

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ABSTRACT

As joint operations have become a key trend in modern military development, unmanned aerial vehicles (UAVs) play an increasingly important role in enhancing the intelligence and responsiveness of combat systems. However, the heterogeneity of aircraft, partial observability, and dynamic uncertainty in operational airspace pose significant challenges to autonomous collision avoidance using traditional methods. To address these issues, this paper proposes an adaptive collision avoidance approach for UAVs based on deep reinforcement learning. First, a unified uncertainty model incorporating dynamic wind fields is constructed to capture the complexity of joint operational environments. Then, to effectively handle the heterogeneity between manned and unmanned aircraft and the limitations of dynamic observations, a sector-based partial observation mechanism is designed. A Dynamic Threat Prioritization Assessment algorithm is also proposed to evaluate potential collision threats from multiple dimensions, including time to closest approach, minimum separation distance, and aircraft type. Furthermore, a Hierarchical Prioritized Experience Replay (HPER) mechanism is introduced, which classifies experience samples into high, medium, and low priority levels to preferentially sample critical experiences, thereby improving learning efficiency and accelerating policy convergence. Simulation results show that the proposed HPER-D3QN algorithm outperforms existing methods in terms of learning speed, environmental adaptability, and robustness, significantly enhancing collision avoidance performance and convergence rate. Finally, transfer experiments on a high-fidelity battlefield airspace simulation platform validate the proposed method's deployment potential and practical applicability in complex, real-world joint operational scenarios.

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1. Introduction

With advancements in modern warfare and the rapid evolution of emerging technologies, unmanned aerial vehicles (UAVs), equipped with advanced autonomous decision-making capabilities, have become crucial in enhancing the intelligence and responsiveness of joint operational systems, significantly improving battlefield efficiency [1–3]. However, in real joint operation airspaces, UAVs face a highly complex operational environment. On one hand, the uneven levels of digital infrastructure across different military branches or combat units, along

with the lack of unified information standards, prevent UAVs from obtaining a comprehensive situational awareness of the airspace in real time via ground systems. On the other hand, the battlefield airspace is often characterized by high dynamism and heterogeneous aircraft, which further exacerbates the uncertainty under conditions of limited local perception [4,5]. In such typical partially observable battlefield environments, UAVs must rely on onboard sensors to gather information about their surrounding airspace and autonomously carry out conflict detection and avoidance [6,7]. Therefore, UAVs urgently require real-time, adaptive, and highly robust autonomous collision avoidance decision-making mechanisms to ensure both operational safety and mission effectiveness in joint combat scenarios.

Extensive research efforts have been devoted to addressing the traditional UAV collision avoidance problem, both domestically and internationally. Existing approaches can be broadly

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categorized into three main types. The first category comprises geometry-based methods, including Geometric Collision Detection [8], the Safety Zone Method [9], and Dubins Paths [10]. These methods are generally characterized by their simplicity and low computational cost. However, they often lack adaptability in highly dynamic or multi-threat environments [11]. The second category encompasses heuristic optimization approaches, represented by Genetic Algorithm [12], Particle Swarm Optimization [13], and Ant Colony Optimization [14]. These techniques are more suitable for complex and uncertain scenarios but tend to suffer from slow convergence or premature convergence to local optima, particularly in multi-agent or high-dimensional threat spaces [15]. The third category includes optimization and control-based algorithms, such as Adaptive Dynamic Programming [16], Mixed Integer Linear Programming (MILP) [17], and Model Predictive Control (MPC) [18]. While ADP and MPC can achieve higher decision precision, ADP typically assumes accurate or known system dynamics and relies heavily on model fidelity, which limits its applicability in partially observable or highly uncertain environments. The inherent limitations of these methods make them less effective in complex, dynamic, and heterogeneous battlefield scenarios, thereby underscoring the urgent need for more flexible, efficient, and adaptive collision avoidance strategies for autonomous UAVs.

In recent years, Deep Reinforcement Learning (DRL) has emerged as a prominent research frontier in artificial intelligence and has been extensively applied across various domains, including air combat decision-making [19,20], autonomous driving [21], UAV obstacle avoidance and path planning [22,23], as well as air traffic management [24]. By formulating the UAV collision avoidance problem as a Markov Decision Process (MDP), DRL enables UAVs to learn and optimize collision avoidance strategies through trial-and-error interactions with the environment, thereby facilitating real-time maneuver decision-making in dynamic and uncertain scenarios [25,26]. Yang et al. [19] proposed an autonomous maneuver decision model based on deep Q network (DQN), combined with a one-on-one short-range air combat evaluation model, to achieve autonomous decision-making in high dimensional state and action space. Choo et al. [27] applied DQN to design a basic UAV collision avoidance system, while Li [28] further improved the DQN structure to enhance avoidance performance in multi-UAV settings. Cone et al. [29] introduced an agent-level optimizer to refine the reward function of DQN, leading to better decision quality and higher avoidance success rates, which can partly support future capability development for collision avoidance. While these works demonstrate the potential of DRL in autonomous airspace decision-making, the environments adopted are often overly simplified, such as assuming limited aircraft types, static or low-density threat models, and unrestricted sensing.

On the basis of the above, Zhang et al. [30] extended the DQN framework by introducing Double DQN (DDQN) to resolve cascading conflicts in denser airspace, but still limited its application to homogeneous scenarios, where all agents are either manned or unmanned with identical dynamics and behaviors. Kong et al. [31] designed a novel autonomous aerial combat maneuver strategy generation algorithm with high-performance and high-robustness based on state-adversarial deep deterministic policy gradient algorithm, which also only considered homogeneous scenarios. Chen et al. [32,33] proposed a Rainbow DQN method that accounts for uncertainty factors such as aircraft heading and speed, demonstrating excellent generalization under various levels of uncertainty and airspace density. Nevertheless, this method failed to address dynamic environmental factors, such as wind fields, and does not consider the varying sensitivities of

heterogeneous aircraft to these uncertainties. Zou et al. [34] introduced the proximal policy optimization algorithm based on a differentially flat model for dynamic soaring trajectory planning and optimization in the battlefield confrontations, but the real-time dynamic navigation control effect was not ideal. In summary, most existing DRL-based UAV collision avoidance studies are constrained by simplified environments, homogeneous agent assumptions, and limited observation models, making them insufficiently adaptable to heterogeneous, contested, and dynamically uncertain airspace as encountered in real-world combat scenarios.

In addition, addressing the challenges of poor convergence and slow training speed in uncertain environments, recent works have focused on optimizing the sampling probability of critical experiences within the replay buffer. Li et al. [35] introduced a Temporal-Difference error-based experience replay mechanism, where the error of each experience is calculated as the difference between the action's output value and its estimated value. The sampling probability for each experience was then determined based on its Temporal-Difference error. Sun et al. [36] devised a deep deterministic policy gradient method based on classification experience replay, which calculates the experience priority coefficient and calculate this value for each experience. Wang et al. [7] proposed a Replay Memory Data Deposit Mechanism, which categorizes experiences according to task scenarios to train agents with enhanced performance. This approach allowed for the sampling of experiences across different scenarios, thereby improving both training efficiency and strategy performance. However, while the above methods categorize experiences, they do not fully consider the Temporal-Difference error of each experience, which limits their effectiveness in dynamic and highly uncertain combat environments.

Motivated by the problems above, this paper proposes a novel adaptive collision avoidance method based on deep reinforcement learning to tackle the challenges of UAV collision avoidance in joint operational airspace environments involving manned and unmanned aircraft. The main contributions of this paper are as follows:

- (1) We establish a joint airspace simulation environment that integrates both manned and unmanned aircraft within the shared airspace. Additionally, we introduce an uncertain joint error model for dynamic wind fields to enhance the model's adaptability in highly uncertain environments.
- (2) We employ sector-based partial observation model and Dynamic Threat Prioritization Assessment (DTPA) to expand the joint operational airspace scenarios. The DTPA algorithm simultaneously accounts for time to closest approach, distance at closest approach, and aircraft type, significantly improving the accuracy of intruder threat assessment within each detection sector.
- (3) We propose a Hierarchical Prioritized Experience Replay (HPER) mechanism to manage and sample experiences across three layers during the collision avoidance process. This not only accelerates the convergence speed but also improves the overall performance of the model.

The remainder of this paper is organized as follows: Section 2 presents the primary collision avoidance problem and formulates its mathematical modeling, accompanied by key assumptions and definitions. Section 3 describes the proposed solution framework in detail. Section 4 presents simulation experiments and discusses the results. Finally, Section 5 concludes the study.

2. UAV collision avoidance problem in joint operational airspace

In this section, we first provide a precise definition of the UAV collision avoidance problem within joint operational airspace scenarios to delineate the scope of this study. We then propose a deep reinforcement learning-based framework specifically designed for UAV adaptive collision avoidance with manned and unmanned aircraft in joint operational environments. This framework transforms the UAV collision avoidance problem into an adaptive learning process, enabling effective collision avoidance in uncertain airspace scenarios.

2.1. Problem description

This study focuses on a joint operational airspace scenario, where real-time state information sharing across different operational units is not available. As a result, UAVs cannot obtain timely situational awareness through the command and control system regarding the positions, intentions, or mission statuses of other airborne platforms, such as manned aircraft or other UAVs. Additionally, UAVs are subject to further challenges, including potential command and control link failures and disturbances caused by complex, dynamic environmental conditions. Under these constraints, UAVs are required to operate in a partially observable environment, where they must independently avoid collisions with heterogeneous aircraft and navigate toward mission targets without full knowledge of the airspace.

As illustrated in Fig. 1, the red zones represent the protected zones of battlefield aircraft, while the orange zones indicate warning areas. During flight, the UAV must autonomously and continuously adjust its heading and velocity in real time, based on the type, relative position, and motion dynamics of nearby aircraft detected within its sensing range, as well as the effects of dynamic wind fields. These real-time adjustments are critical to avoiding potential conflicts and ensuring mission success in uncertain joint operational airspace.

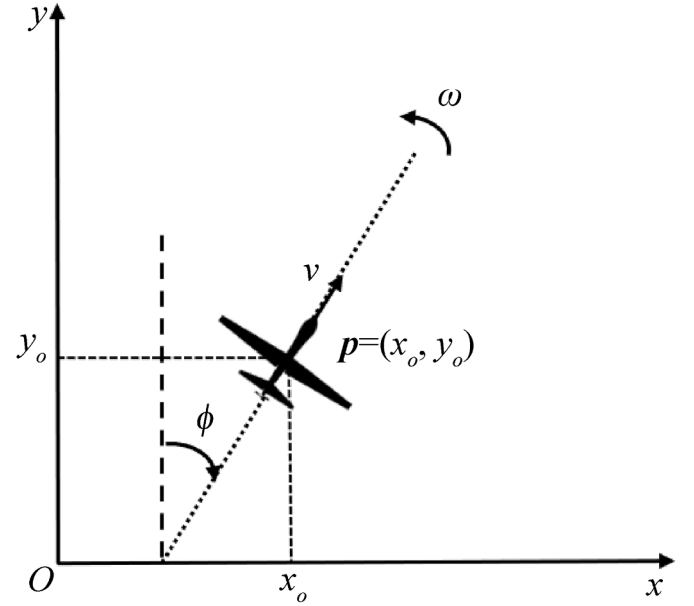


Fig. 2. Schematic of the UAV coordinate system.

2.2. Kinematic model

The second-order motion equations of the UAV are established as follows:

$$\begin{cases} \dot{v} = a_v \\ \dot{x} = v \cos \phi \\ \dot{y} = v \sin \phi \\ \dot{\phi} = \omega \end{cases} \quad (1)$$

where $p(x_o, y_o)$ denotes the position of the UAV in the two-dimensional plane, while v , ϕ , and ω represent the speed, heading angle, and angular velocity of the UAV, respectively. And a_v denotes the linear acceleration. Fig. 2 shows the coordinate system of the UAV.

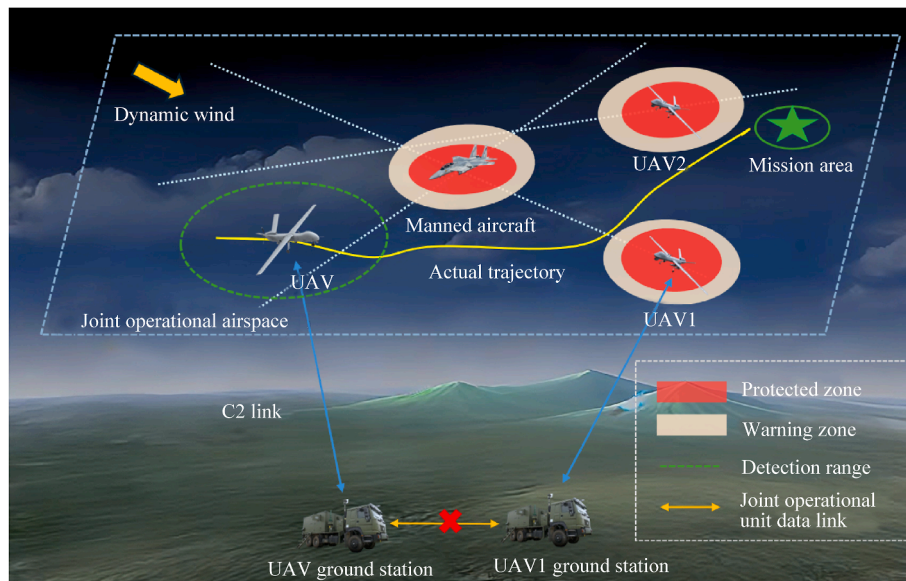


Fig. 1. Diagram of UAV collision avoidance joint airspace scenarios.

2.3. Learning environment

The collision avoidance task for UAVs in joint operational airspace requires support for highly dynamic and uncertain environments. This paper designs and develops a simulation platform for manned and unmanned aircraft integrated operational scenarios, with the following main functionalities: (1) providing the agent with rich state information to support real-time decision-making; (2) evaluating the collision avoidance strategies of the agent; and (3) providing reward feedback for the agent's actions. This simulation platform offers UAVs a realistic dynamic environment, enabling them to perceive conflicts and progressively optimize their collision avoidance capabilities.

2.3.1. Joint operational airspace with manned and unmanned aircraft

This study defines a contested airspace scenario for joint manned and unmanned aircraft operations. In this scenario, N_m manned aircraft and N_u unmanned aircraft operate in the same airspace at the same altitude. Additionally, to analyze the impact of dynamic wind fields on manned and unmanned aircraft, a dynamic wind field is incorporated into the environment, where wind speed and wind direction are represented by v_w and ϕ_w , respectively.

The positions and initializations of manned aircraft and unmanned aircraft are generated based on their respective motion capabilities and in accordance with predefined combat plans. It is worth noting that the manned and unmanned aircraft in the environment do not strictly follow linear flight paths but are influenced by uncertainty factors (see Section 2.2.2 for details). From Fig. 3(a), a simulated environment containing 10 aircraft, with the image showing a 2D projection of the horizontal plane.

To enhance safety, this paper proposes a dual-layer safety protection interval model, which includes the protected zone and the warning zone. The radii of these two zones are defined as R_p and R_w , respectively. A collision is considered to occur when the distance between UAV and other aircraft is less than R_p . A warning signal is triggered when entering the warning zone. Since manned aircraft typically have higher speeds and longer reaction times, their warning zone is usually greater than that of UAVs.

The relationship between the UAV and manned/unmanned aircraft in the environment is depicted in Fig. 3(b). Taking the relationship between the UAV and a manned aircraft as an example, the Closest Point of Approach (CPA) is the point of closest proximity in the horizontal plane, with D_{CPA} representing the

distance at the CPA, and T_{CPA} representing the time to reach the CPA. The relative distance $\mathbf{r}_{rel}$ and relative velocity $\mathbf{v}_{rel}$ between the agent and the manned aircraft are given by:

$$\begin{cases} \mathbf{r}_{rel} = \mathbf{p}_{int} - \mathbf{p}_{own} \\ \mathbf{v}_{rel} = \mathbf{v}_{int} - \mathbf{v}_{own} \end{cases} \quad (2)$$

where $\mathbf{p}_{int}$ and $\mathbf{p}_{own}$ denote the position vectors of the intruder aircraft and the ownship UAV, respectively, while $\mathbf{v}_{int}$ and $\mathbf{v}_{own}$ represent their respective velocity vectors.

The time and distance at the CPA are calculated by:

$$\begin{cases} T_{CPA} = \frac{\mathbf{r}_{rel} \cdot \mathbf{r}_{rel}}{\mathbf{v}_{rel} \cdot \mathbf{v}_{rel}} \\ D_{CPA} = \sqrt{\mathbf{r}_{rel} \cdot \mathbf{r}_{rel} - \frac{(\mathbf{r}_{rel} \cdot \mathbf{v}_{rel})^2}{\mathbf{v}_{rel} \cdot \mathbf{v}_{rel}}} \end{cases} \quad (3)$$

These metrics, D_{CPA} and T_{CPA} , are crucial for assessing the collision risk between the UAV and other aircraft during flight. In Section 3.2, these metrics will be further utilized to evaluate the threat levels posed by surrounding aircraft to the agent.

2.3.2. Environment uncertain model

In joint operational airspace, the dynamic wind field significantly impacts the performance of UAVs and manned aircraft, serving as a key factor contributing to environmental uncertainty. To reflect the periodic variations of wind speed and direction observed in real-world airspace, this study adopts a periodic dynamic wind field model. The wind field speed v_w and direction ϕ_w are defined as:

$$\begin{cases} v_w(t) = v_{base} + v_{var} \sin\left(\frac{2\pi t}{T}\right) \\ \phi_w(t) = \phi_{base} + \phi_{var} \sin\left(\frac{2\pi t}{T}\right) \end{cases} \quad (4)$$

where v_{base} and v_{var} represent the baseline value and fluctuation amplitude of the wind speed, respectively, while ϕ_{base} and ϕ_{var} represent the baseline value and fluctuation amplitude of the wind direction, respectively. T denotes the period of wind speed variation.

To mirror the distinct sensitivities of UAV and manned aircraft to wind fields, this study introduces a wind field sensitivity weighting coefficient η^{type} , which delineates the degree of wind field influence on different types of aircraft. Generally, UAVs are

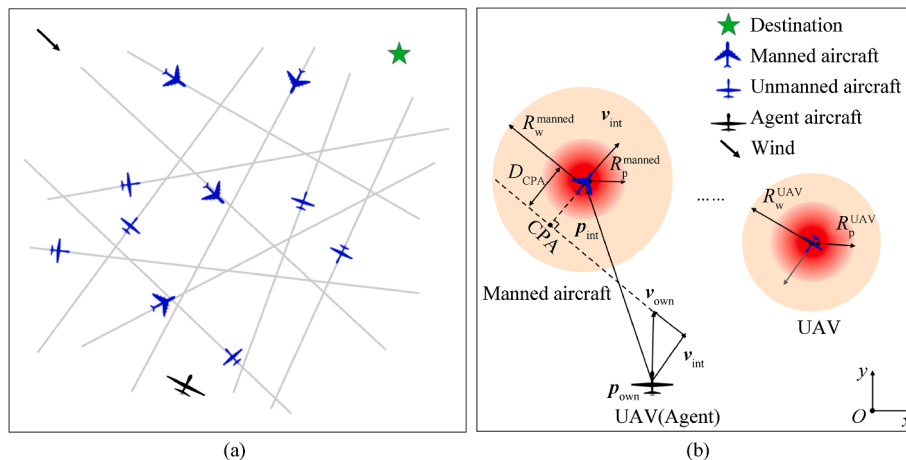


Fig. 3. UAV collision avoidance with manned/unmanned aircraft. (a) An example of joint operational airspace scenarios; (b) Illustration of the conflict relationship.

more sensitive to wind disturbances, characterized by $\eta^{UAV} > \eta^{manned}$, indicating that UAVs are more significantly affected by wind field perturbations.

Under the influence of dynamic wind fields, the ground speed $\mathbf{v}_s$ of an aircraft is the vector sum of its original speed $\mathbf{v}_o$ and the wind field speed $\mathbf{v}_w$, as illustrated in Fig. 4. The ground speed for a given type of aircraft is defined as:

$$\mathbf{v}_s^{type} = \mathbf{v}_o + \eta^{type} \mathbf{v}_w \quad (5)$$

The magnitude of the ground speed v_s^{type} and its heading ϕ_s^{type} are further defined as follows:

$$\begin{cases} v_s^{type} = \sqrt{(v_{s,x}^{type})^2 + (v_{s,y}^{type})^2} \\ \phi_s^{type} = \arctan\left(\frac{v_y^{type}}{v_x^{type}}\right) \end{cases} \quad (6)$$

In addition to wind field disturbances, the uncertainty in an aircraft's flight state (including heading angle and speed) significantly impacts its accuracy. These uncertainties primarily stem from flight control system errors, external disturbances, and sensor measurement inaccuracies. To model these effects, this study represents the heading angle error σ_ϕ and speed error σ_v as random variables following a standard normal distribution, with their ranges determined by the 95% confidence interval. The error ranges for UAVs and manned aircraft are defined as:

$$\begin{cases} \sigma_\phi^{manned} \in [-B_\phi^{manned}, B_\phi^{manned}], \sigma_\phi^{UAV} \in [-B_\phi^{UAV}, B_\phi^{UAV}] \\ \sigma_v^{manned} \in [-B_v^{manned}, B_v^{manned}], \sigma_v^{UAV} \in [-B_v^{UAV}, B_v^{UAV}] \end{cases} \quad (7)$$

Here, $B_v^{UAV} > B_v^{manned}$ and $B_\phi^{UAV} > B_\phi^{manned}$, indicating that UAVs are more susceptible to external environmental disturbances compared to manned aircraft. Incorporating these uncertainties, the actual speed and heading angle of the aircraft are expressed as:

$$\begin{cases} v_s^{type} = v_s^{type} + \sigma_v^{type} \\ \phi_s^{type} = \phi_s^{type} + \sigma_\phi^{type} \end{cases} \quad (8)$$

This study comprehensively considers the complexity and uncertainty of aircraft operations by integrating the uncertainties of

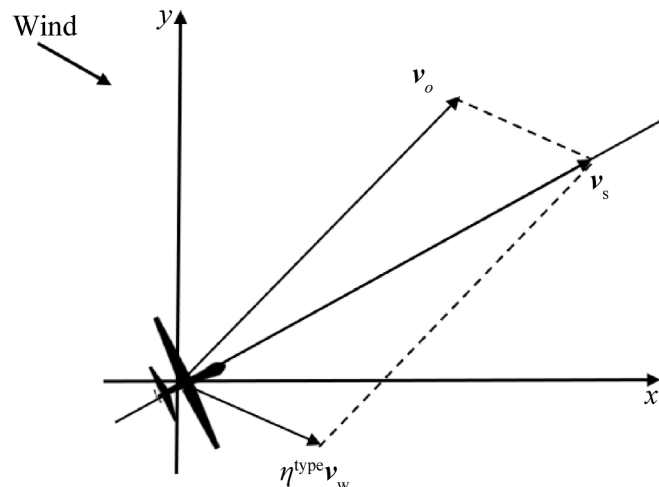


Fig. 4. Dynamic model of wind field.

the wind field and flight state. This joint error model provides a realistic foundation for simulating the flight states of manned and unmanned aircraft joint operation airspace environments, providing a reliable basis for the development and validation of flight control and collision avoidance algorithms in complex scenarios.

2.3.3. Observation space

In an ideal scenario, the observation space would encompass the global state information of all aircraft, enabling the agent to achieve comprehensive situational awareness within its detection range. However, it is difficult to obtain global information in the battlefield environment, and global observations are often incompatible with fixed-dimension observation matrices. Additionally, relying entirely on global information is constrained by real battlefield airspace environmental limitations and can significantly increase computational resource consumption. We adopt a sector-based partial observation method to enhance information acquisition efficiency and reduce computational complexity, which is depicted in Fig. 5.

The agent (i.e., the ownship) has a detection radius R_d , and the detection range is divided into K sectors, starting from the north direction and proceeding clockwise. Aircraft within the detection range are referred to as "detected aircraft". To effectively assess the potential threats posed by detected aircraft, this study introduces a DTPA based on partial observation information. The detailed algorithm is provided in Section 3. DTPA evaluates the threats posed by both manned and unmanned aircraft within the detected sectors, identifies the most threatening aircraft in each sector, and marks it as the "intruder" for that sector.

Based on this design, the partial observation state matrix of the agent at time t , denoted as $\mathbf{O}^t$, consists of the ownship's observation state $\mathbf{O}_0^t$ and the intruder's observation state $\mathbf{O}_1^t$, as follows:

$$\mathbf{O}^t = [\mathbf{O}_0^t, \mathbf{O}_1^t] \quad (9)$$

The intruder's observation state $\mathbf{O}_1^t$ is composed of the observation states of intruders in the K detected sectors:

$$\mathbf{O}_1^t = [\mathbf{O}_1^t, \mathbf{O}_2^t, \dots, \mathbf{O}_d^t, \dots, \mathbf{O}_K^t] \quad (10)$$

where $\mathbf{O}_d^t$ represents the observation matrix for the intruder in the d th detection sector, as shown in Fig. 6.

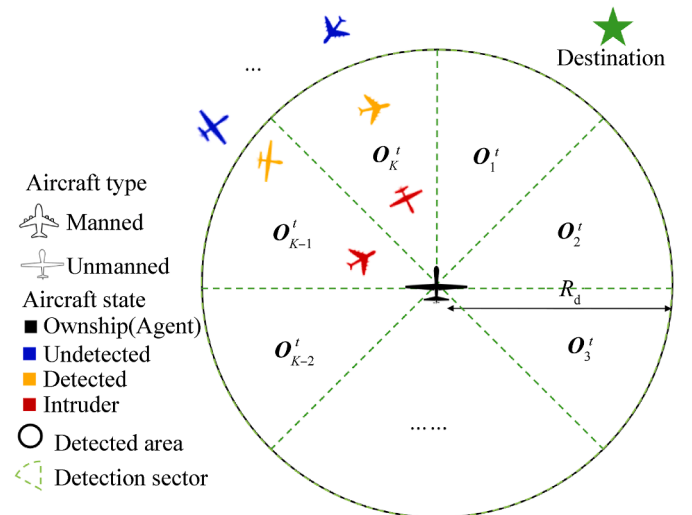


Fig. 5. Sector-Based partial observation model.

(1) Observation state of the ownship

The partial observation state of the ownship $\mathbf{o}_O^t$ at time t , includes its velocity $\mathbf{o}_{ov}^t$, destination point information $\mathbf{o}_{og}^t$, and environmental vector $\mathbf{o}_{oe}^t$. It is defined as:

$$\mathbf{o}_O^t = [\mathbf{o}_{ov}^t, \mathbf{o}_{og}^t, \mathbf{o}_{oe}^t] \quad (11)$$

(A) Velocity vector $\mathbf{o}_{ov}^t$: The velocity vector of the ownship aircraft consists of its current speed:

$$\mathbf{o}_{ov}^t = [v_{own}^t] \quad (12)$$

(B) Mission destination position vector $\mathbf{o}_{og}^t$: The velocity provides key information about the relative relationship between the UAV and the mission destination point. It includes the distance to the destination point D_g^t and the heading angle toward the destination point ϕ_g^t :

$$\begin{cases} \mathbf{o}_{og}^t = [D_g^t, \phi_g^t] \\ D_g^t = \|\mathbf{p}_g^t - \mathbf{p}_{own}^t\| \\ \phi_g^t = \arccos\left(\frac{(\mathbf{p}_g^t - \mathbf{p}_{own}^t) \cdot \mathbf{v}_{own}^t}{\|\mathbf{p}_g^t - \mathbf{p}_{own}^t\| \|\mathbf{v}_{own}^t\|}\right) \end{cases} \quad (13)$$

where $\mathbf{p}_g^t$ and $\mathbf{p}_{own}^t$ represent the position vectors of the destination point and the ownship at time t , respectively.

(C) Environmental vector $\mathbf{o}_{ow}^t$: This vector provides wind field information, including wind speed and wind direction:

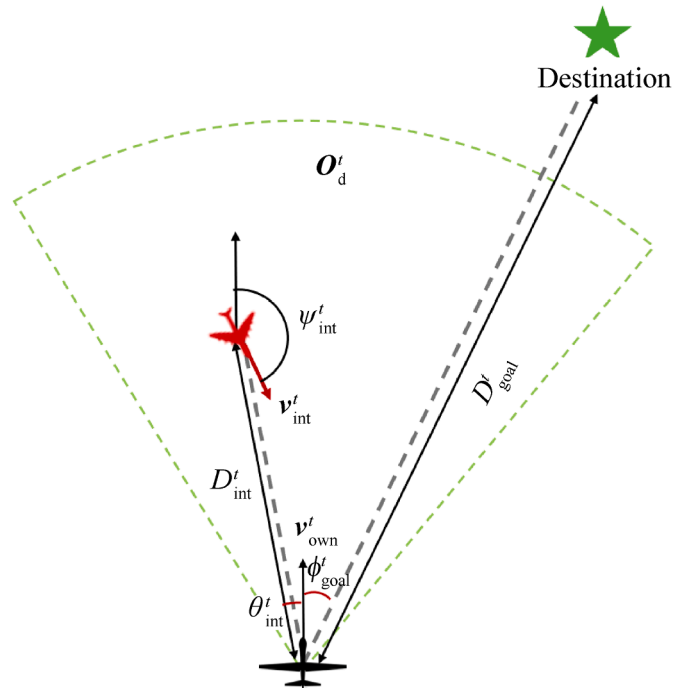


Fig. 6. Observation sector define.

$$\mathbf{o}_{ow}^t = [v_w^t, \phi_w^t] \quad (14)$$

(2) Observation state of the intruder

The observation state matrix $\mathbf{o}_d^t$ for the d th detection sector includes the positional relationship $\mathbf{o}_{ip}^t$, type $\mathbf{o}_{it}^t$, and velocity of the intruder aircraft $\mathbf{o}_{iv}^t$:

$$\mathbf{o}_d^t = [\mathbf{o}_{ip}^t, \mathbf{o}_{it}^t, \mathbf{o}_{iv}^t] \quad (15)$$

(A) Intruder position vector $\mathbf{o}_{ip}^t$: This vector describes the relative relationship between the ownship and the intruder, evaluating the threat posed by the intruder. It includes the distance to the intruder D_{int}^t , the relative heading angle of the intruder θ_{int}^t , and the angle between the heading of the intruder and the ownship ψ_{int}^t . It is calculated by:

$$\begin{cases} \mathbf{o}_{ip}^t = [D_{int}^t, \theta_{int}^t, \psi_{int}^t] \\ D_{int}^t = \|\mathbf{p}_{int}^t - \mathbf{p}_{own}^t\| \\ \theta_{int}^t = \arccos\left(\frac{(\mathbf{p}_{int}^t - \mathbf{p}_{own}^t) \cdot \mathbf{v}_{own}^t}{\|\mathbf{p}_{int}^t - \mathbf{p}_{own}^t\| \|\mathbf{v}_{own}^t\|}\right) \\ \psi_{int}^t = \arccos\left(\frac{\mathbf{v}_{int}^t \cdot \mathbf{v}_{own}^t}{\|\mathbf{v}_{int}^t\| \|\mathbf{v}_{own}^t\|}\right) \end{cases} \quad (16)$$

(B) Type vector $\mathbf{o}_{it}^t$: This vector indicates the type of intruder aircraft, such as manned aircraft or unmanned aircraft:

$$\mathbf{o}_{it}^t = \begin{cases} [1] & \text{if type}_{int} \text{ is manned aircraft} \\ [0] & \text{if type}_{int} \text{ is unmanned aircraft} \end{cases} \quad (17)$$

(C) Velocity vector $\mathbf{o}_{iv}^t$: The velocity vector of the intruder includes its speed:

$$\mathbf{o}_{iv}^t = [v_{int}^t] \quad (18)$$

Through this design of the partially observation space, the agent can focus on handling high-threat situations within a local range while ensuring real-time performance and reducing computational complexity. This approach enhances the collision avoidance and adaptability of the agent in manned and unmanned aircraft joint operation airspace environments.

2.3.4. Action space

In battlefield airspace environments, the design of the action space is one of the core elements for UAVs to achieve efficient collision avoidance. In this study, the action space is designed as a discrete set of two-dimensional maneuver action, each representing a combination of linear acceleration a_v and angular velocity a_ω . Specifically, $a_v \in \{-a_{adj}, 0, -a_{adj}\}$ and $a_\omega \in \{-\omega_{adj}, 0, +\omega_{adj}\}$, forming a total of 9 discrete action combinations.

At each time step, the UAV's speed v_{own}^t and heading angle ϕ_{own}^t updated based on the selected action a^t . The update process is defined as:

$$\begin{cases} v_{\text{own}}^{t+1} = \text{Clip}(v_{\text{own}}^t + a_v \cdot \Delta t) \\ \phi_{\text{own}}^{t+1} = \phi_{\text{own}}^t + a_\omega \cdot \Delta t \end{cases} \quad (19)$$

2.3.5. Reward function

In joint operational airspace environments, to ensure that UAV complete mission safely and effectively in complex and dynamic conditions, this study designs a multi-factor reward function. This reward function comprehensively incorporates factors such as reaching the destination, avoiding collisions, preventing boundary violations, and maintaining flight efficiency. It encourages the UAV to learn effective collision avoidance strategies by providing feedback for its actions at each time step. Specifically, the reward function r^t at time t is defined as follows:

- (1) Mission destination reward: A positive reward is provided when the UAV successfully reaches the combat mission destination, defined as the distance between the UAV and the destination point D_g being smaller than the destination radius R_g . This encourages the UAV reach destination quickly. The mission destination reward at t time step is defined as:

$$r_{\text{goal}}^t = \begin{cases} C_{\text{goal}} & D_g^t < R_g \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

where C_{goal} is the positive reward coefficient assigned for reaching the destination.

- (2) Warning penalty reward: The penalty applies a negative reward when the UAV enters the warning zone of another aircraft. This penalty encourages the UAV to exit high-risk areas promptly. Given the varying characteristics of manned aircraft and UAV, this study adopts an adaptive penalty mechanism that adjusts the threshold and penalty intensity based on the intruder type. The warning penalty reward at time step t is defined as:

$$r_{\text{w}}^t = \begin{cases} C_{\text{w}} & D_{\text{int}}^t < R_{\text{w}}^{\text{type}} \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

where C_{c} is the negative penalty coefficient for collision events.

- (3) Collision penalty reward: A significant negative reward is imposed when a collision occurs, which is defined as the situation where the distance D_{int}^t between the UAV and the intruder is less than the radius of the protected zone R_p . The collision penalty reward at time step t is defined as:

$$r_{\text{c}}^t = \begin{cases} C_{\text{c}} & D_{\text{int}}^t < R_p \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

where C_{c} is the negative penalty coefficient for collision events.

- (4) Boundary violation penalty reward: A negative reward is applied if the UAV leaves the predefined airspace. This ensures the UAV operates within the designated area to avoid potential safety risks. The boundary penalty reward at times step t is defined as:

$$r_{\text{out}}^t = \begin{cases} C_{\text{out}} & \text{UAV is out of bounds} \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

where C_{out} is the negative penalty coefficient for going out of bounds.

- (5) Efficiency reward: To encourage efficient flight toward the destination point, a negative reward is applied at each time step. Additionally, this reward considers the change in distance to the target and the angular deviation between the current heading and the destination direction. The efficiency reward at time step t is defined as:

$$r_{\text{e}}^t = C_{\text{step}} + C_{\text{dis}}(D_g^t - D_g^{t-1}) + C_{\text{angle}}|\phi_g^t| \quad (24)$$

where C_{step} is the base penalty per time step, C_{dis} and C_{angle} are coefficients for distance change and angular deviation.

The total reward function r_{total}^t at time step t integrates the aforementioned five reward components to balance safety and efficiency, as computed in Eq. (25). The weights for different rewards are designed based on their importance. Due to the critical nature of constraints, such as, aircraft collision, entry into warning zones, boundary violations, and reaching the target point, larger weights are assigned to these rewards, with $C_{\text{goal}} = 2$, $C_{\text{out}} = -0.5$, $C_{\text{c}} = -1$, and $C_{\text{w}} = -0.5$. On the other hand, the efficiency reward is treated as a soft constraint and is assigned smaller weights, with $C_{\text{step}} = -0.005$, $C_{\text{dis}} = -0.001$, and $C_{\text{angle}} = -0.001$.

$$r_{\text{total}}^t = r_{\text{goal}}^t + r_{\text{ca}}^t + r_{\text{w}}^t + r_{\text{out}}^t + r_{\text{e}}^t \quad (25)$$

3. Proposed method

In the joint operational airspace environment investigated in this study, the environment is characterized by high dynamism and uncertainty. Traditional DQN encounters significant challenges in capturing the complexity of such dynamic environments, particularly due to the limitations of their experience replay mechanisms, which can lead to degraded performance or even failure. To address this issue, we propose an enhanced experience replay strategy—namely, the HPER mechanism.

This section begins with a brief overview to the foundational frameworks of DQN and Double-Dueling Deep Q-Network (D3QN). Section 3.2 and Section 3.3 then describe the DTPA and the HPER mechanism, respectively. Finally, Section 3.4 outlines the overall framework of the proposed HPER-D3QN method, including pseudocode for its training process. The complete methodological framework is illustrated in Fig. 7.

3.1. DQN theoretical approach

DQN [37] is the classic deep-reinforcement-learning framework that combines Q-learning, deep neural networks, and experience replay to solve high-dimensional decision-making problems. It approximates the action-value function $Q(\mathbf{O}, \alpha; \theta)$, thereby estimating the value of each state-action pair.

We adopt D3QN as the foundation of our approach [38], which integrates the DDQN update rule [39] with the dueling network architecture [40]. By maintaining two networks—an online (prediction) network for action selection and a target network for value evaluation—D3QN reduces Q-value over-estimation; the

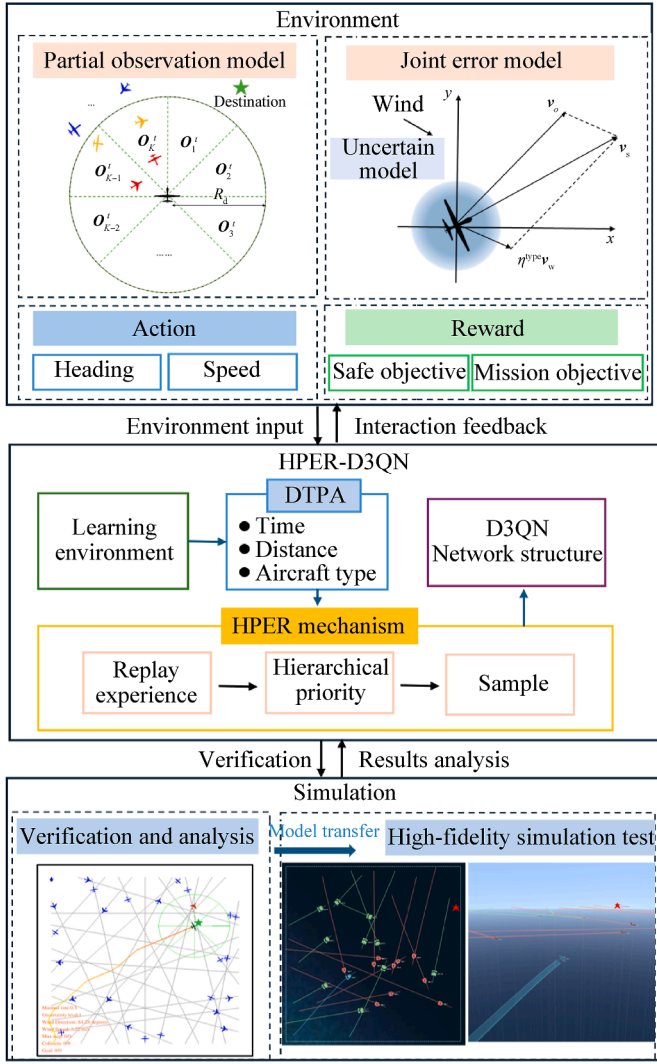


Fig. 7. Framework of UAV collision avoidance method based on HPER-D3QN.

target update formula is defined as:

$$y = r + \gamma Q(\mathbf{O}', \arg\max_a Q(\mathbf{O}', a'; \theta); \theta^-) \quad (26)$$

where r is the immediate reward, γ is the discount factor, $\mathbf{O}'$ is the next observation, a' is the next action, θ are the parameters of the online network, and θ^- are the parameters of the target network.

In addition, the dueling design decomposes Q into a state-value branch $Q(\mathbf{O})$ and an action-advantage branch $A(\mathbf{O}, \alpha; \theta, \alpha)$; their combination, shown in Eq. (27), yields more accurate state-value estimates:

$$Q(\mathbf{O}, a; \theta, \alpha, \beta) = V(\mathbf{O}; \theta, \beta) + \left(A(\mathbf{O}, \alpha; \theta, \alpha) - \frac{1}{|A|} \sum_{a'} A(\mathbf{O}, a'; \theta, \alpha) \right) \quad (27)$$

Here, $V(\mathbf{O}; \theta, \beta)$ is the state value function and $A(\mathbf{O}, \alpha; \theta, \alpha)$ the action advantage function. The dueling architecture allows the model to accurately evaluate states even when the effects of all possible actions are similar.

3.2. DTPA mechanism

In most existing learning-based methods [30,41,42], threat prioritization is typically determined based solely on the Euclidean distance between the UAV and nearby detected aircraft. However, these approach fails to fully capture the actual severity of threats in dynamic environments, especially when dealing with heterogeneous aircraft types and time-varying wind disturbances. For example, when encountering a high-speed manned aircraft, relying solely on distance may underestimate the true collision risk.

To address this limitation, we propose a novel DTPA mechanism that incorporates multi-dimensional information—including time to closest approach T_{CPA} , distance at closest approach D_{CPA} , and aircraft type—to dynamically adjust threat priorities. This design significantly improves the accuracy of threat assessment and enhances the adaptability of decision-making under complex and uncertain conditions. The core idea of the DTPA is to calculate the threat priority for each aircraft within the UAV's detection sector and identify the highest-priority aircraft as the intruder. The main steps of the algorithm are as follows:

- (1) Calculate T_{CPA} and D_{CPA} between the UAV and each detected aircraft within the detection sector. The specific calculation methods are provided in Eq. (3).
- (2) Determine the priority coefficient based on the type of detected aircraft, accounting for differences in their operational characteristics.
- (3) Evaluate the threat score for each detected aircraft using a weighted multi-factor formula:

$$S = \omega_t \frac{T_{CPA} - T_{\min}}{T_{\max} - T_{\min}} + \omega_d \frac{D_{CPA} - D_{\min}}{D_{\max} - D_{\min}} + \omega_{\text{type}} \kappa \quad (28)$$

where $T_{\max}$ and $T_{\min}$ represent the maximum and minimum time thresholds for the UAV's closest approach to another aircraft, respectively. Similarly, $D_{\max}$ and $D_{\min}$ correspond to the maximum and minimum distance thresholds. The weight coefficients ω_t , ω_d , and ω_{type} are assigned to the time, distance, and aircraft type, respectively, to prioritize these factors based on their importance in collision avoidance. Additionally, κ is a factor that adjusts the score according to the type of aircraft involved in the potential threat.

- (4) Rank the detected aircraft based on their calculated threat scores and select the aircraft with the highest threat score as the intruder within the detection sector.

Table 1
Experience feature parameters.

Parameters	Description	Value	
		Yes	No
e_d	UAV detects other aircraft	1	0
e_w	UAV enters another aircraft's warning zone	1	0
e_c	UAV enters the protected zone or not	1	0
e_{out}	UAV exceeds airspace boundaries	1	0
e_g	UAV arrives destination point	1	0

airspace, priority is given to avoiding manned aircraft. Thus, when encountering manned and unmanned aircraft, the values of κ are set to 0.75 and 0.25, respectively.

Algorithm 1. Dynamic Threat Prioritization Assessment Algorithm.

1.	Input: detected aircraft in detection sectors
2.	Output: intruder of the detection sectors
3.	Initialize: $\omega_t, \omega_d, \omega_{type}, \kappa, S \leftarrow \emptyset$
4.	$T_{max}, T_{min}, D_{max}, D_{min}$
5.	for aircraft i in detected aircraft do :
6.	Computer T_{CPA} and D_{CPA} using Eq. (3)
7.	Computer threat score S_i using Eq. (28)
8.	Append S_i to S
9.	end for
10.	Sort S in descending order
11.	Select the intruder with the highest S

3.3. HPER mechanism

Traditional experience replay mechanisms sample experience transitions uniformly at random, disregarding the critical importance of certain samples for policy learning, leading to low training efficiency. To address this limitation, we propose a novel HPER mechanism, specifically designed to enhance learning efficiency in joint operational scenarios.

In the studied environment, three key characteristics are identified: (1) the environment is highly uncertain; (2) collision avoidance priorities differ between manned and unmanned aircraft, and (3) the UAV collision avoidance process involves multiple contextual scenarios, such as entering warning zones, breaching minimum protected distances, reaching the destination, or exceeding the airspace boundaries, as described in Section 3.2. These characteristics lead to the generation of diverse types of experiences during the agent's training process, each with varying importance for policy optimization. As a result, HPER is introduced to categorize experiences into hierarchical layers based on their contribution to strategy optimization. By managing and sampling experiences from different layers, HPER enhances training efficiency and improves policy performance.

In joint operational environments, the experience replay buffer $\mathbf{E}$ is represented as:

$$\mathbf{E} = \{\mathbf{e}_i | \mathbf{e}_i = (\mathbf{O}_i, a_i, \mathbf{O}'_i, r_i), i = 1, 2, \dots, M\} \quad (29)$$

where each experience $\mathbf{e}_i$ consists of the observation $\mathbf{O}_i$, action a_i , next observation $\mathbf{O}'_i$, and reward r_i . M denotes the maximum storage capacity of the experience replay buffer.

Based on the analysis of the observation states, the UAV collision avoidance process in joint operational environments can be quantitatively represented using a feature vector. The feature vector is defined as:

$$\|\mathbf{O}_i\| = [e_d, e_w, e_c, e_{out}, e_g] \quad (30)$$

where $\|\mathbf{O}_i\|$ represents the contextualized observation state, e_d indicates whether the UAV detects other aircraft, e_w indicates whether the UAV has entered the warning zone of other aircraft, e_c indicates whether the UAV has entered the protected zone of other aircraft, e_{out} indicates whether the UAV has exceeded the defined airspace boundaries, e_g indicates whether the UAV has reached the target point. The values and definitions of $e_d, e_w, e_c, e_{out}, e_g$ are listed in Table 1.

Based on the above definitions, the observation state $\mathbf{O}$ of the UAV during the collision avoidance process can be categorized by several distinct scenarios, $\mathbf{O}_{safe}$ indicates UAV detects no aircraft in

the detection sector. $\mathbf{O}_w$ indicates UAV enters the warning zone of another aircraft. $\mathbf{O}_c$ indicates UAV enters the minimum protected distance of another aircraft. $\mathbf{O}_{out}$ indicates UAV exceeds the airspace boundary. $\mathbf{O}_{arrival}$ indicates UAV arrives destination point. These scenarios are mathematically described as Eqs. (31) and (32), and the corresponding observation scenario parameters are summarized in Table 2.

$$\mathbf{O} = \{\mathbf{O}_{safe}, \mathbf{O}_w, \mathbf{O}_c, \mathbf{O}_{out}, \mathbf{O}_{arrival}\} \quad (31)$$

$$\mathbf{O}_{type} = \{\mathbf{O}_i | \|\mathbf{O}_i\| = [e_d, e_w, e_c, e_{out}, e_{arrival}]\} \quad (32)$$

The experience replay buffer $\mathbf{E}$ is divided into three priority layers: high-priority experience layer $\mathbf{E}_{high}$, medium-priority experience layer $\mathbf{E}_{medium}$, and low-priority experience layer $\mathbf{E}_{low}$. Each layer is assigned a fixed storage capacity, and when the storage limit is reached, old experiences are removed using a first-in-first-out [43] strategy. This approach ensures that more critical experiences are retained for training, while less important ones are discarded over time, improving the efficiency of the learning process.

(1) High-priority experience layer

The high-priority experience layer includes experiences related to reaching the destination, collisions, and airspace violations. It is defined as:

$$\mathbf{E}_{high} = \mathbf{E}_{arrival} \cup \mathbf{E}_{collision} \cup \mathbf{E}_{out} \quad (33)$$

$$\mathbf{E}_{arrival} = \{\mathbf{e}_i | \mathbf{O}_i \in \mathbf{O}_{safe}, \mathbf{O}'_i \in \mathbf{O}_{arrival} \cup \mathbf{O}_i \in \mathbf{O}_w, \mathbf{O}'_i \in \mathbf{O}_{arrival}\} \quad (34)$$

$$\mathbf{E}_{out} = \{\mathbf{e}_i | \mathbf{O}_i \in \mathbf{O}_{safe}, \mathbf{O}'_i \in \mathbf{O}_{out} \cup \mathbf{O}_i \in \mathbf{O}_w, \mathbf{O}'_i \in \mathbf{O}_{out}\} \quad (35)$$

$$\mathbf{E}_{collision} = \{\mathbf{e}_i | \mathbf{O}_i \in \mathbf{O}_{safe}, \mathbf{O}'_i \in \mathbf{O}_c \cup \mathbf{O}_i \in \mathbf{O}_w, \mathbf{O}'_i \in \mathbf{O}_c\} \quad (36)$$

These terminal states directly influence policy optimization objectives. Successfully reaching the destination provides positive reinforcement, enhancing the agent's task achievement capabilities, while collision or airspace violation events offer negative feedback, steering the strategy away from unsafe behaviors. Although these experiences are rare, they carry high value and are crucial for effective learning. As such, they are prioritized during sampling to ensure the agent can adequately adapt to these critical scenarios and improve its decision-making process in high-risk situations.

(2) Medium-priority experience layer

The medium-priority experience layer includes scenarios where the UAV enters the warning zone of another aircraft:

Table 2
Observation scenario parameters.

Observation type	Observation parameters values				
	e_d	e_w	e_c	e_{out}	e_g
$\mathbf{O}_{safe}$	0	0	0	0	0
$\mathbf{O}_w$	1	1	0	0	0
$\mathbf{O}_c$	1	1	1	0	0
$\mathbf{O}_{out}$	{0,1}	{0,1}	0	1	0
$\mathbf{O}_{arrival}$	{0,1}	{0,1}	0	0	1

$$\mathbf{E}_{\text{medium}} = \{ \mathbf{e}_i | \mathbf{O}_i \in \mathbf{O}_{\text{safe}}, \mathbf{O}'_i \in \mathbf{O}_w \cup \mathbf{O}_i \in \mathbf{O}_w, \mathbf{O}'_i \in \mathbf{O}_w \} \quad (37)$$

These experiences signify potential risks, allowing the model to learn threat perception and early avoidance strategies. While not terminal states, they are vital for refining the UAV's ability to adapt to dynamic threats. Therefore, these experiences are prioritized less than high-priority experiences but more than routine experiences, striking a balance between sampling frequency and decision optimization.

(3) Low-priority experience layer

The low-priority experience layer includes scenarios where the UAV detects no aircraft in the detection area:

$$\mathbf{E}_{\text{low}} = \{ \mathbf{e}_i | \mathbf{O}_i \in \mathbf{O}_{\text{safe}}, \mathbf{O}'_i \in \mathbf{O}_{\text{safe}} \} \quad (38)$$

These experiences are primarily used to maintain the UAV's autonomous flight capabilities. However, compared to terminal and warning scenarios, they have a lesser impact on policy optimization. Assigning them a lower priority prevents them from diluting the sampling of more valuable experiences, accelerating strategy convergence.

During training, the HPER mechanism determines the sampling weight λ_{layer} for each layer as:

$$\lambda_{\text{layer}} = \frac{M_{\text{layer}}}{M} k_{\text{layer}} \quad (39)$$

where k_{layer} is the priority scaling factor, M_{layer} is the number of experiences in the layer, and M is the total number of experience across all layers. and p_{layer} denotes the sampling probability of the corresponding layer. Training process mini-batch size is m . The number of samples m_{layer} from each layer is then calculated as:

$$m_{\text{layer}} = \frac{\lambda_{\text{layer}}}{\sum_{\text{layer}} p_{\text{layer}}} m \quad (40)$$

Furthermore, within each layer, the sampling probability for an individual experience i is determined using it Temporal-Difference [44] error δ_i :

$$\delta_i = r_i + \gamma Q(\mathbf{O}'_i, \mathbf{a}'_i; \theta) - Q(\mathbf{O}_i, \mathbf{a}_i; \theta) \quad (41)$$

$$p_{\text{layer}}(i) = \frac{|\delta_i|^{\alpha_{\text{PER}}} + \epsilon}{\sum_{j \in \text{layer}} (|\delta_j|^{\alpha_{\text{PER}}} + \epsilon)} \quad (42)$$

where α_{PER} controls the importance of the δ_i , and ϵ is a smoothing term to prevent zero probabilities. Here, δ_j denotes the Temporal-Difference error of the j -th experience in the layer. This hierarchical and prioritized sampling approach ensures the model focuses on critical scenarios, accelerates convergence, and improves policy robustness in complex joint operational airspace. Algorithm 2 outlines the detailed HPER operation process.

Algorithm 2. HPER mechanism.

1. **Input:** Experience replay buffer $\mathbf{E}$, buffer sizes M , priority scaling factors $k_{\text{high}}, k_{\text{medium}}, k_{\text{low}}$, batch size m
2. **Output:** Mini-batch for training
3. **Initialize:**
Divide $\mathbf{E}$ into layers: $\mathbf{E}_{\text{high}}, \mathbf{E}_{\text{medium}}, \mathbf{E}_{\text{low}}$
Assign fixed capacities: $M_{\text{high}}, M_{\text{medium}}, M_{\text{low}}$
4. **Store Experience:**
5. **for** each experience $\mathbf{e}_i = (\mathbf{O}_i, \mathbf{a}_i, \mathbf{O}'_i, r_i)$ **do:**

(continued on next page)

(continued)

6. **if** $\mathbf{O}_i \in \{\mathbf{O}_{\text{arrival}}, \mathbf{O}_c, \mathbf{O}_{\text{out}}\}$:
7. Add $\mathbf{e}_i$ to $\mathbf{E}_{\text{high}}$
8. **else if** $\mathbf{O}_i \in \{\mathbf{O}_w\}$
9. Add $\mathbf{e}_i$ to $\mathbf{E}_{\text{medium}}$
10. **else:**
11. Add $\mathbf{e}_i$ to $\mathbf{E}_{\text{low}}$
12. **end for**
13. **Sample Experience:** Compute sampling weight for each layer use Eq. (39)
14. **for** each layer $\mathbf{E}_{\text{layer}}$:
15. Compute TD-error δ_i use Eq. (41)
16. Compute sampling probability for each experience use Eq. (42)
17. Sample m_{layer} experiences using p_{layer}
18. **end for**
19. Combine experiences for all layers to form the mini-batch
20. Return: Mini-batch for training

3.4. HPER-D3QN method

The workflow of the agent is illustrated in Fig. 8 and Algorithm 3. Each training episode in the proposed algorithm consists of two stages: data generation and collection, and policy update. During the data generation and collection stage, the simulation scenario is first reset (Line 5), and the time step is initialized (Line 6). At each time step, the agent obtains the current local observation $\mathbf{O}^t$ through Algorithm 1 (Line 9), computes the Q-values for all possible actions using the prediction network, and selects an avoidance action $\mathbf{a}^t$ using the ϵ -greedy policy (Line 10). The selected action is then executed to receive the immediate reward r^t and observe the next state $\mathbf{O}^{t+1}$ (Line 11). This forms a transition sample $(\mathbf{O}^t, \mathbf{a}^t, r^t, \mathbf{O}^{t+1})$, which is assigned a priority based on the TD error and stored into the corresponding priority layer of the experience replay buffer (Line 13).

In the policy update stage, the HPER mechanism samples a fixed number m of mini-batch samples from the high, medium, and low-priority experience buffers respectively (Line 14). The DDQN strategy is then used to compute the target Q-values (Line 15), and the parameters of the prediction network are updated based on the TD loss function, completing one full training iteration.

Algorithm 3. HPER-D3QN Algorithm.

1. **Input:** The dataset of simulation scenarios
2. **Output:** parameters of target network and prediction network θ
3. **Initialize:**
Experience replay buffer $\mathbf{E}_{\text{high}}, \mathbf{E}_{\text{high}}, \mathbf{E}_{\text{high}}$ for high/medium/low priority, evaluation network parameter θ and target network parameter $\bar{\theta}$, and set hyper-parameters as show in Table 4
4. **for** episode = 1, 2, 3 ..., **do:**
5. Reset the simulation scenario
6. $t \leftarrow 0$
7. **While** the UAV has not collision, not reached destination, not out of bounds **do**
8. Get the observation $\mathbf{O}^t$ using Algorithm 1
9. Evaluate the $Q(\mathbf{O}^t, \mathbf{a}; \theta)$, using Eq. (27)
10. Choose $\mathbf{a}_t \sim \epsilon$ -greedy(Q)
11. Execute $\mathbf{a}_t$, obtain r_t , observe $\mathbf{O}^{t+1}$
12. Form experience $\mathbf{e}_t = (\mathbf{O}^t, \mathbf{a}_t, \mathbf{O}^{t+1}, r_t)$
13. Store $\mathbf{e}_t \rightarrow \mathbf{E}_{\text{high}}/\mathbf{E}_{\text{medium}}/\mathbf{E}_{\text{low}}$
14. Sample m mini-batch experiences for each priority layer, using Eq. (40)
15. Calculate DDQN target
16. Update θ via TD loss
17. Every C step: $\bar{\theta} \leftarrow \theta$
18. $t \leftarrow t + 1$
19. **end while**
20. **end for**

4. Simulation and discussion

In this section, we first describe the experimental setup,

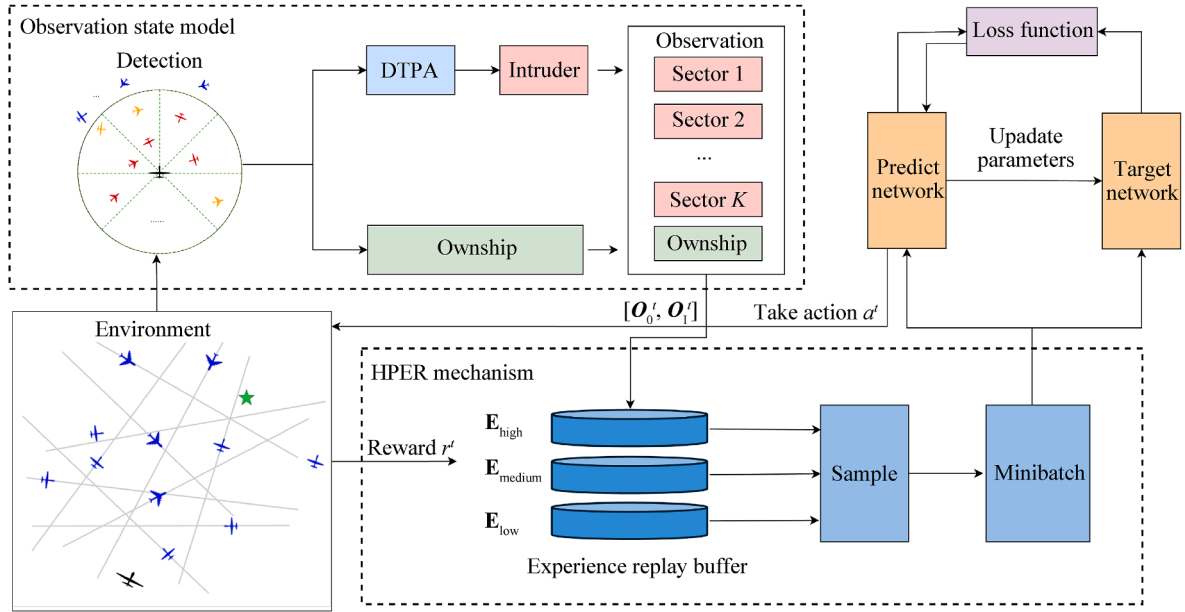


Fig. 8. Structure and data flow of the HPER-D3QN.

including the generation of the training scenario. Section 4.2 presents the training process of the agent, while Section 4.3 discusses the testing of the agent under various conditions.

Table 3
Parameter settings for training scenarios.

Parameters	Values
Number of aircraft	20
The number ratio of UAV	0.5
Time of each time step	1s
UAV speed range	[60,110] m·s ⁻¹
Manned aircraft speed range	[80,160] m·s ⁻¹
Radius of the destination	100 m
Radius of the protected zone	150 m
Radius of the UAV warning zone	400 m
Radius of the manned aircraft warning zone	800 m
Radius of the agent detection area	4000 m
Number of detection sectors	8
Uncertainty of environment	3 level
Speed adjusts	5 m·s ⁻¹
Heading adjusts	3°·s ⁻¹
$\omega_t, \omega_d, \omega_{type}$	0.4, 0.4, 0.2
$\eta_{UAV}, \eta_{manned}$	1.0, 0.8
Maximum and minimum time thresholds	20 s, 200 s
Maximum and minimum distance thresholds	150 m, 4000 m

Table 4
Hyper-parameter settings for training algorithm.

Hyperparameter	Value
Network structure	45×128×64×32×9
Minibatch size m	128
Experiences replay buffer size	20000
High, medium, low layer replay buffer size	[4000,6000,10000]
Learning rate α	0.0001
Discount factor γ	0.99
Greedy rate ϵ	0.995 → 0.001
The priority scaling factor of each layer	[1–3]
Controls the prioritization of experiences	0.5
Target network update cycle	50

4.1. Environment setup

This study utilizes the PyGame library to develop a training environment, as shown in Fig. 3(a), which represents a 30 km × 30 km joint operational airspace. The environment can support up to 25 aircraft (excluding the agent) within the same airspace, with the number of UAV being adjustable to simulate different scenarios in the airspace. The manned and unmanned aircraft follow predefined flight combat plan. Specific environment parameters are detailed in Table 3.

The model is implemented in Python using the PyTorch library and trained on TITAN RTX GPU. The neural network receives an observation matrix as input, with a five-layer network architecture. Detailed hyperparameters for training are provided in Table 4.

4.2. Training results

To validate the effectiveness of the proposed method, we compared the training results of DQN, DDQN, Dueling DQN, D3QN,

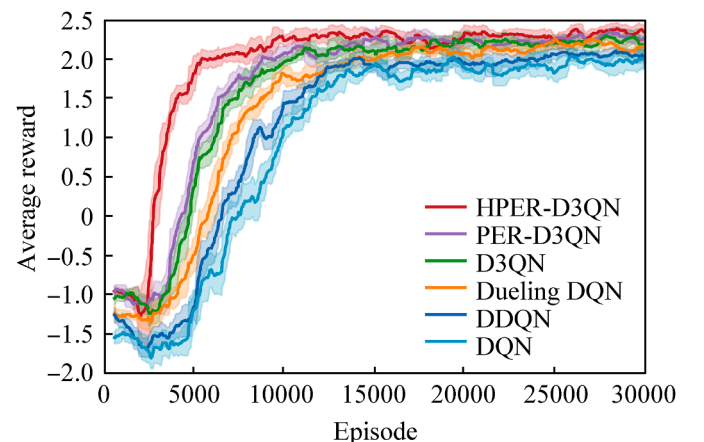


Fig. 9. Training curves.

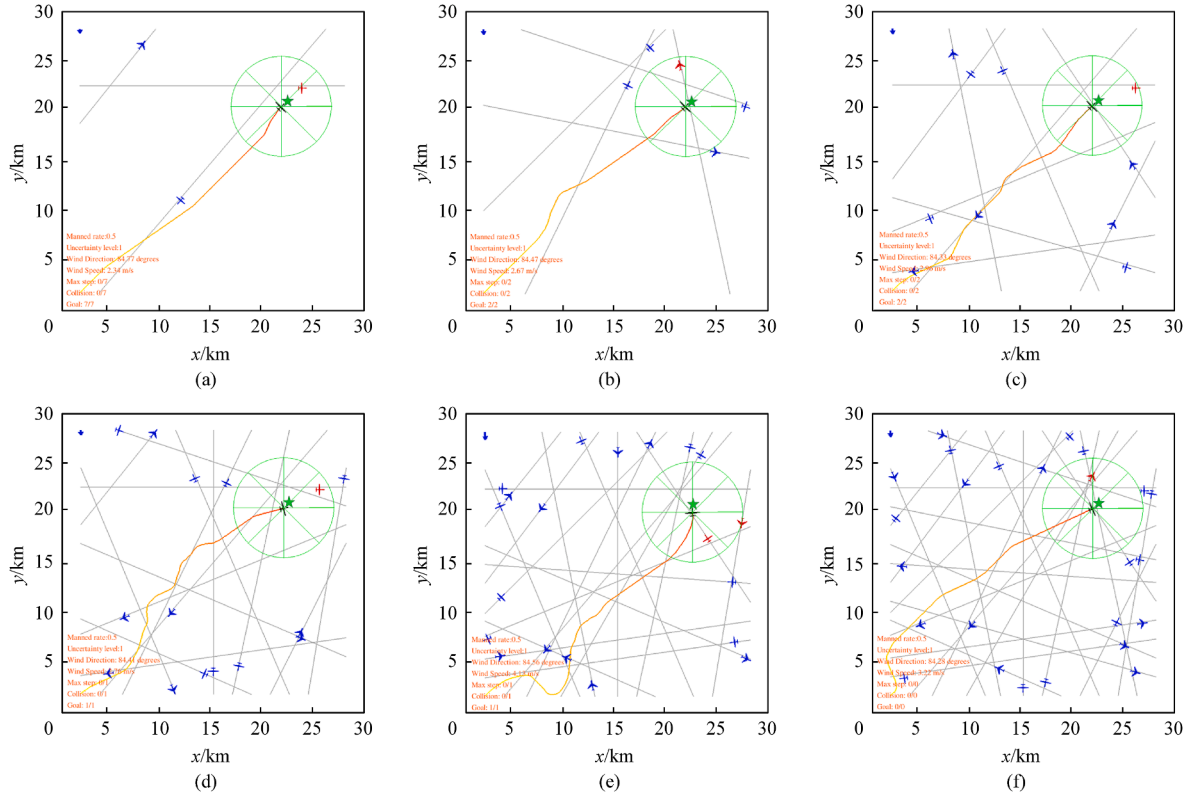


Fig. 10. Test scenarios with different numbers of aircraft. (a) Aircraft number = 3; (b) Aircraft number = 5; (c) Aircraft number = 15; (d) Aircraft number = 15; (e) Aircraft number = 20; (f) Aircraft number = 25.

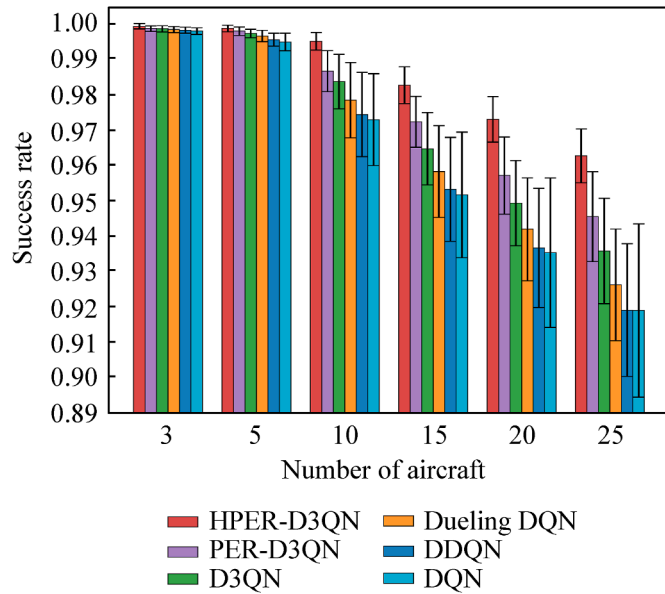


Fig. 11. Success rate with different numbers of aircraft.

PER-D3QN, and HPER-D3QN under the same training hyperparameters and environment. During training, the number of aircraft in each episode, the ratio of manned to unmanned aircraft, and the environmental uncertainty parameters were randomly selected. The average reward values during training for each method were recorded, as shown in Fig. 9.

To better visualize the variations during training, we applied a

sliding window smoothing technique to the actual reward values. The solid line in the figure represents the smoothed average reward values, while the transparent area indicates the standard deviation of the average reward. As observed in Fig. 9, all methods experienced a decline in reward values in the early stages of training due to frequent collisions and boundary violations during the exploration phase. After around 3500 episodes, the reward curves for all methods began to rise, with HPER-D3QN showing the fastest growth rate. Furthermore, from the standard deviation area in Fig. 9, it was evident that HPER-D3QN exhibited relatively smaller fluctuations, indicating a more stable training process. After around 25,000 episodes, all methods began to gradually converge. The reward value for HPER-D3QN converged to 2.36, which was consistently higher than that of the other methods, demonstrating superior performance and faster convergence. Next, the agents trained with these methods are tested in the evaluation environment.

4.3. Performance test

In this section, we evaluated the performance of the agent trained using the proposed HPER-D3QN method. To assess the effectiveness of the HPER mechanism, we also conducted tests on DQN, DDQN, Dueling DQN, D3QN, and PER-DQN under the same environment. The performance comparisons were conducted from three perspectives: the number of aircraft in the environment, the level of environmental uncertainty, and the ratio of manned to unmanned aircraft.

4.3.1. Testing with different numbers of aircraft

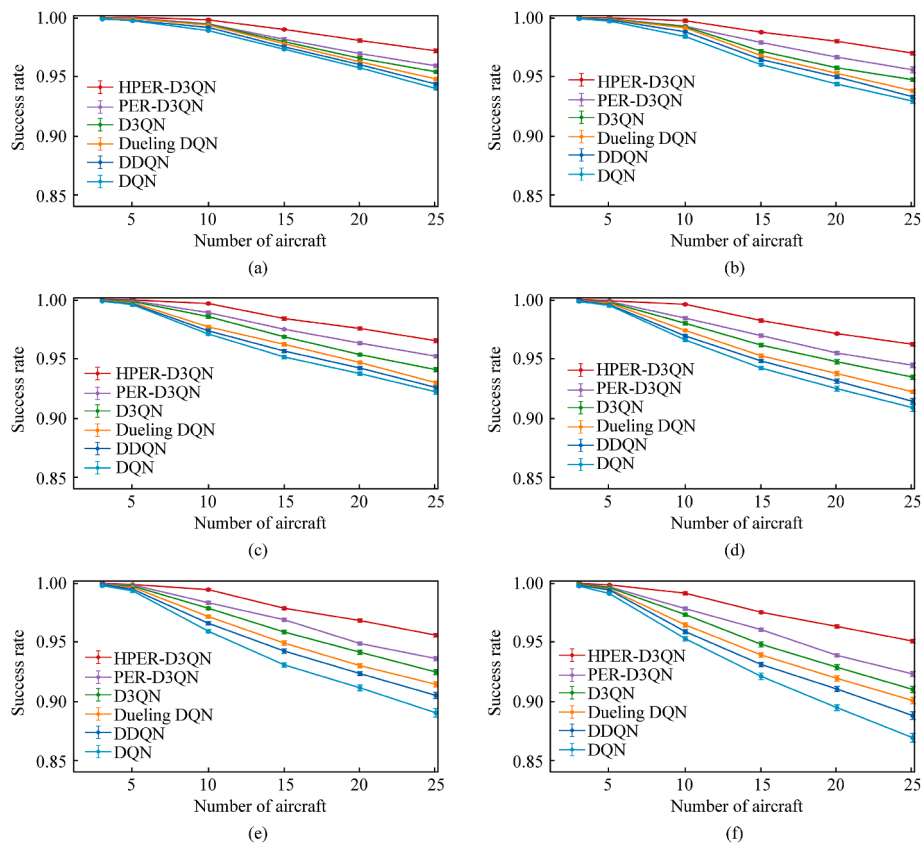
To evaluate the performance of the trained agent under different aircraft densities, we constructed a series of randomized

Table 5
Metrics with different numbers of aircraft.

Method	Success rate/%			Task Completion Time/s			FHP/(times/episode) $\times 10^{-2}$		
	$N = 3$	$N = 25$	Δ_S	$N = 3$	$N = 25$	Δ_T	$N = 3$	$N = 25$	Δ_F
HPER-D3QN	99.95	96.28	−3.67	293.26	336.17	42.91	0.08	4.92	4.84
PER-D3QN	99.89	94.55	−5.34	292.37	347.56	55.19	0.26	7.16	6.90
D3QN	99.88	93.58	−6.30	295.82	346.81	51.01	0.34	7.82	7.48
Dueling DQN	99.86	92.61	−7.25	297.89	353.79	55.90	0.42	8.56	8.14
DDQN	99.80	91.90	−7.90	298.24	361.17	62.93	0.41	8.52	8.11
DQN	99.80	91.84	−7.96	298.86	364.81	65.95	0.41	8.64	8.23

Table 6
Level of uncertainty.

Uncertainty Level l	Manned aircraft		Unmanned aircraft		Wind			
	Heading error/ ($^{\circ}$)	Speed error/ ($\text{m}\cdot\text{s}^{-1}$)	Heading error/ ($^{\circ}$)	Speed error/ ($\text{m}\cdot\text{s}^{-1}$)	Direction/ ($^{\circ}$)	Speed/ ($\text{m}\cdot\text{s}^{-1}$)	Direction Fluctuation/ ($^{\circ}$)	Speed Fluctuation/ ($\text{m}\cdot\text{s}^{-1}$)
0	0	0	0	0	[0,90,180,270]	0	0	0
1	[−1,1]	[−0.5,0.5]	[−2,2]	[−1,1]	[0,90,180,270]	4	1	1
2	[−2,2]	[−1,1]	[−4,4]	[−2,2]	[0,90,180,270]	6	2	2
3	[−3,3]	[−1.5,1.5]	[−6,6]	[−3,3]	[0,90,180,270]	8	3	3
4	[−4,4]	[−2,2]	[−8,8]	[−4,4]	[0,90,180,270]	10	4	4
5	[−5,5]	[−2.5,2.5]	[−10,10]	[−5,5]	[0,90,180,270]	12	5	5

**Fig. 12.** Success rate with different uncertainty levels. (a) Uncertainty level 0; (b) Uncertainty level 1; (c) Uncertainty level 2; (d) Uncertainty level 3; (e) Uncertainty level 4; (f) Uncertainty level 5.

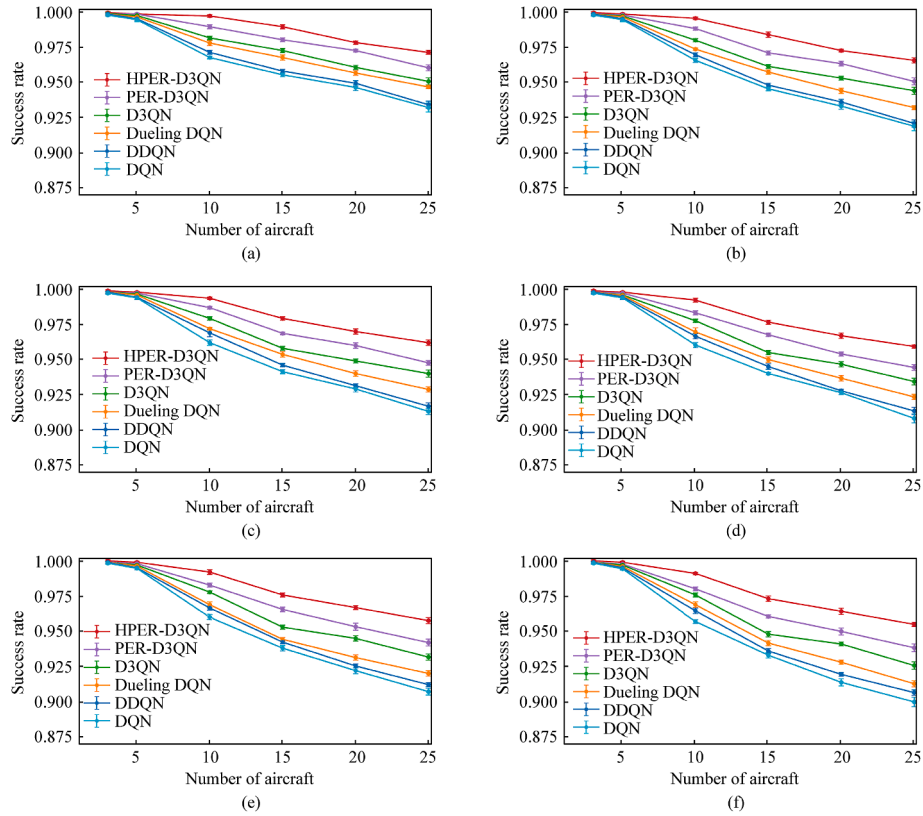
joint operation airspace scenarios with varying numbers of aircraft: 3, 5, 10, 15, 20, and 25. The randomness of each scenario was reflected in the variation of initial positions, destinations, and dynamic uncertainties during flight. Fig. 10 illustrated representative flight trajectories under different numbers of aircraft.

For each traffic density, we conducted 300 independent

simulation runs, repeated 10 times. The performance evaluation metrics include: (1) Success rate, defined as the proportion of episodes in which the agent successfully avoids collisions and reaches its destination; (2) Task completion time, referring to the average time taken by the agent to complete its mission; and (3) Frequency of Hazardous Proximity (FHP), which measures the

Table 7
Metrics with different uncertainty levels.

Method	Success rate/%			Task Completion Time/s			FHP/(times/episode) $\times 10^{-2}$		
	$l = 0$	$l = 5$	Δ_S	$l = 0$	$l = 5$	Δ_T	$l = 0$	$l = 5$	Δ_F
HPER-D3QN	97.16	95.06	−1.56	328.21	342.57	14.36	3.46	7.26	3.80
PER-D3QN	95.90	92.3	−3.60	331.64	349.63	17.99	5.92	9.53	3.61
D3QN	95.40	91.00	−4.40	334.29	351.84	17.55	6.69	9.84	3.15
Dueling DQN	94.80	90.06	−4.20	335.73	359.18	23.45	7.98	12.53	4.55
DDQN	94.36	88.80	−5.56	337.60	364.25	26.65	8.16	12.52	4.36
DQN	94	86.93	−7.07	339.62	364.17	24.55	8.19	14.89	6.70

**Fig. 13.** Success rate with different ratios. (a) The ratio of UAV = 0; (b) The ratio of UAV = 0.2; (c) The ratio of UAV = 0.4; (d) The ratio of UAV = 0.6; (e) The ratio of UAV = 0.8; (f) The ratio of UAV = 1.0.

average number of times per episode the agent enters the warning zone of another aircraft.

Fig. 11 presented the success rates and standard deviations for all methods under varying traffic densities. It could be observed that when the number of aircraft is low (3 or 5), achieve success rates close to 100%. However, as the number of aircraft increased, the success rates began to decline, with the DQN method showing the steepest drop. In contrast, HPER-D3QN exhibited the smallest

degradation, maintaining a high success rate of 96.4% even in the dense scenario with 25 aircraft, significantly outperforming the baseline methods.

Table 5 provided further quantitative results comparing task completion time and FHP between low-density ($N = 3$) and high-density ($N = 25$) environments. As expected, both metrics increased with traffic density across all methods. However, HPER-D3QN showed the smallest increases: task completion time

Table 8
Metrics under different manned-unmanned aircraft ratios.

Method	Success rate/%			Task Completion Time/s			FHP (times/episode) $\times 10^{-2}$		
	$\rho = 0$	$\rho = 1$	Δ_S	$\rho = 0$	$\rho = 1$	Δ_T	$\rho = 0$	$\rho = 1$	Δ_F
HPER-D3QN	97.13	95.43	−1.7	339.68	332.51	−7.17	6.68	4.72	−1.96
PER-D3QN	96.03	93.76	−2.27	351.41	339.96	−11.45	8.68	7.62	−1.06
D3QN	95.06	92.50	−2.56	350.86	342.38	−8.48	9.64	8.14	−1.50
Dueling DQN	94.66	91.20	−3.46	358.91	349.98	−8.93	10.48	9.24	−1.24
DDQN	93.40	90.56	−2.84	363.53	351.32	−12.21	11.62	10.23	−1.39
DQN	93.20	89.90	−3.30	362.74	352.56	−10.18	12.89	10.48	−2.41

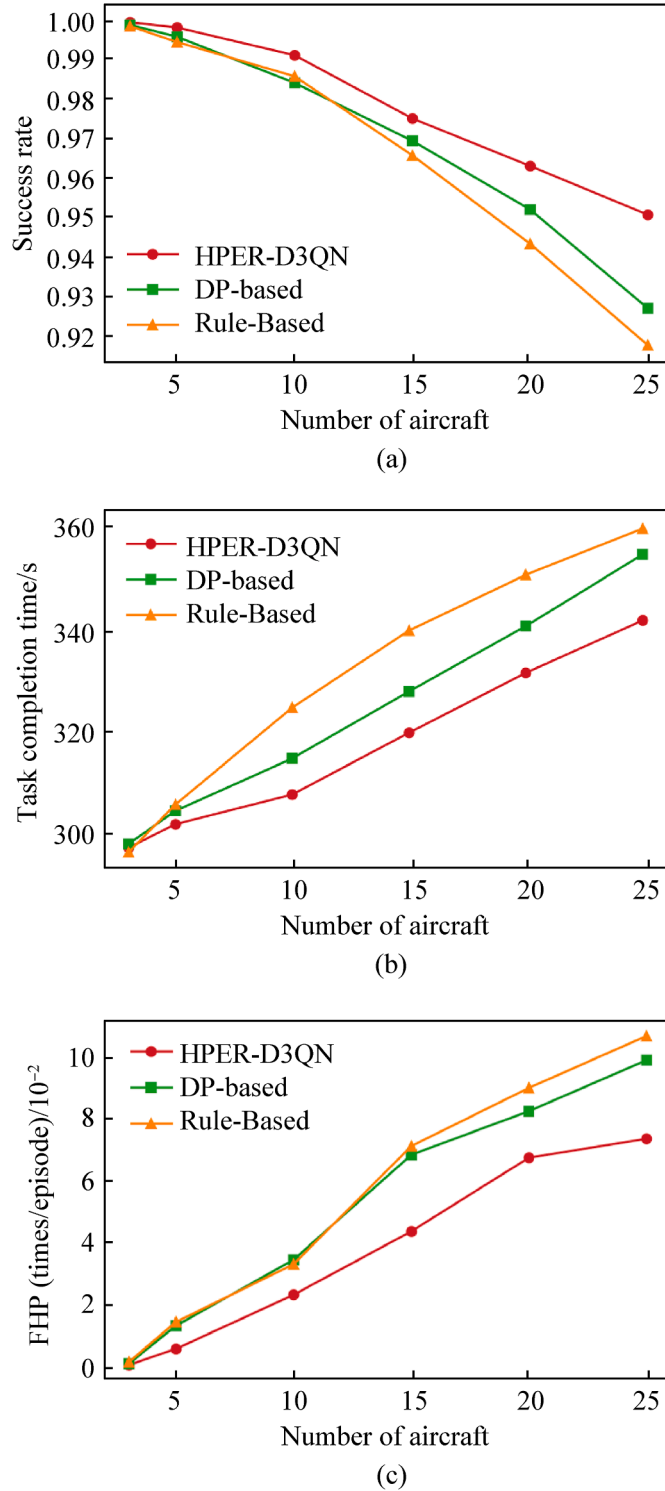


Fig. 14. Compare the results of the three methods. (a) Comparison results of success rates; (b) Comparison results of task completion times; (c) Comparison results of FHP.

increased by only 42.91 s, and FHP rose by just 4.84×10^{-2} . In comparison, the DQN method's task completion time increased by 65.95 s, and FHP increased by 8.23×10^{-2} . These results demonstrated that HPER-D3QN exhibits superior adaptability and robustness in both sparse and congested airspace scenarios.

Table 9

Results of ablation experiments.

Ablation item	Success rate/%		Task Completion Time/s		FHP (times/episode) $\times 10^{-2}$	
	Metrics	Loss	Metrics	Loss/%	Metrics	Loss/%
baseline	95.82	0	344.58	0	7.38	0
No DTPA	87.21	-8.98	357.94	-3.88	14.21	92.54
No HPER	86.93	-9.27	361.31	-4.86	13.82	87.26
No Dueling	91.31	-4.71	350.22	-1.64	9.84	33.33
No Double	92.81	-3.14	351.27	-1.94	9.73	31.84

4.3.2. Testing with different uncertainty levels

To further evaluate the robustness and adaptability of the proposed HPER-D3QN method under varying levels of environmental uncertainty, we designed a set of testing scenarios with graded uncertainty conditions. The uncertainty model was constructed based on recent studies on aircraft error modeling [33,45] and aligned with the operational characteristics of UAV outlined by the International Civil Aviation Organization [46]. Specifically, six levels of uncertainty were defined (as shown in Table 6), incorporating velocity errors, heading angle perturbations, and dynamic wind disturbances. Here, uncertainty level $l = 0$ corresponds to an ideal scenario with no disturbance, while level $l = 5$ represents the most challenging environment with high-intensity perturbations.

For each uncertainty level, we adopted the same testing setup and evaluation metrics as described in Section 4.3.1. As shown in Fig. 12, the results indicated that at all uncertainty levels, as the number of aircraft increases, the success rates of all the methods decrease. However, the success rate of the HPER-D3QN method remained higher than that of other methods. Fig. 12(a) showed that in the environment with no uncertainty, when the number of aircraft reaches its maximum, all the methods maintain a success rate above 94%, with HPER-D3QN achieving 97.16%. In Figs. 12(b)–12(e), we could see that as the uncertainty level increases, the success rates of all the methods decrease, and the rate of decline becomes more significant. Notably, the decline for HPER-D3QN was smaller than for other methods. Fig. 12(f) showed that in the most challenging scenario with uncertainty level $l = 5$, the success rate of DQN drops to 86.93% under the highest aircraft density, whereas HPER-D3QN still maintains a high performance with a success rate of 95.06%.

Beyond success rates, Table 7 further quantified the changes in task completion time and FHP. The results showed that both metrics increase with higher uncertainty levels, indicating that environmental disturbances significantly raise mission complexity and collision risk. Nevertheless, HPER-D3QN consistently yielded the lowest growth in both metrics, demonstrating superior environmental adaptability. Specifically, the task completion time for HPER-D3QN increased modestly from 328.21 s to 342.57 s (an increase of 14.36 s), while the DQN method saw a larger increase from 339.62 s to 364.17 s (an increase of 24.55 s). Regarding FHP, HPER-D3QN increased from 3.46×10^{-2} to 7.26×10^{-2} times per episode, a marginal rise of 3.80×10^{-2} , whereas DQN's FHP grew by 6.70×10^{-2} . Based on the analysis of Fig. 12 and Table 7, it can be concluded that HPER-D3QN demonstrates the best collision avoidance performance and strongest robustness compared to other methods under different uncertainty conditions.

4.3.3. Testing with different manned-unmanned aircraft ratios

To further evaluate the performance of the trained agent under different aircraft type ratios, we designed multiple test environments with different proportions of UAVs. Considering the impact



Fig. 15. Snapshots of the test environment in simulation platform.

of aircraft type and quantity on the agent's collision avoidance strategy in real-world integrated airspace, we set the UAV ratio $\rho \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$. In each case, the number of unmanned aircraft was set to $\lceil \rho N \rceil$, and the number of manned aircraft to $N - \lceil \rho N \rceil$. As the influence of uncertainty has already been discussed in Section 4.3.2, we set the uncertainty level to 2 throughout this experiment to ensure a controlled comparison. The testing methodology and evaluation metrics remained consistent with Section 4.3.1.

As shown in Fig. 13, the success rates of all methods decreased as the unmanned aircraft ratio increased, especially in high-density traffic scenarios. This trend is mainly attributed to two factors: (1) unmanned are modeled with higher uncertainty than manned aircraft, making them more susceptible to environmental disturbances; (2) the larger separation buffer of manned aircraft provides more predictable avoidance cues for the agent. Despite the overall declining trend, HPER-D3QN consistently exhibited superior stability and adaptability across all ratios, maintaining a success rate above 95%, significantly outperforming baseline methods.

Further evidence from Table 8 revealed that HPER-D3QN also outperforms others in task completion time and FHP. Specifically, at $\rho = 0$, HPER-D3QN's task completion time increased by only 7.17 s, while DDQN and DQN exceeded 10 s, indicating HPER-D3QN's stronger adaptability to change in aircraft composition. In terms of FHP, HPER-D3QN showed the smallest increase, from 6.68×10^{-2} to 4.72×10^{-2} times per episode, outperforming all other methods. Notably, in scenarios with a high proportion of manned aircraft, the larger warning zones lead to fewer close encounters, thereby facilitating safer and more efficient agent behavior. In summary, HPER-D3QN demonstrated stable collision avoidance performance across a wide range of manned–unmanned aircraft ratios, achieving higher success rates, shorter task durations, and lower FHP than other comparison methods.

4.3.4. Method comparison

To further assess the effectiveness of the proposed approach, two representative advanced collision avoidance methods were selected as baselines for comparison: (1) the rule-based collision avoidance method developed by NASA (hereinafter referred to as

Rule-Based) [47], and (2) a next-generation UAV collision avoidance method based on dynamic programming [48] (hereinafter referred to as DP-Based). To ensure a fair comparison, all three methods were evaluated under identical simulation conditions, including consistent initial and destination point distributions, a fixed uncertainty level of 5, and uniform airspace configuration and aircraft dynamics parameters.

Fig. 14 presented the comparative performance of the three methods under varying numbers of aircraft. As shown in Fig. 14(a), all methods exhibited a decline in performance as the number of aircraft increases. However, HPER-D3QN consistently maintained a success rate above 95% even under high-density traffic conditions, significantly outperforming both the DP-Based and Rule-Based methods. Regarding task completion time, all methods showed an increasing trend as the number of aircraft rose, due to the more complex conflict resolution required in denser scenarios. Notably, HPER-D3QN achieved the shortest task completion times across all densities, indicating its ability to maintain high efficiency while ensuring safety. In terms of FHP, HPER-D3QN demonstrated the smallest rate of increase, suggesting that it more effectively avoids entering the warning zones of other aircraft and thus reduces potential collision risks. In contrast, the FHP values for the Rule-Based and DP-Based methods increased rapidly with traffic density, indicating a higher likelihood of encountering potential threats in congested airspace. In summary, HPER-D3QN outperformed the benchmark methods across all three key metrics—success rate, task completion time, and FHP—particularly under high-density and complex airspace conditions.

4.3.5. Ablation experiment

To evaluate the contribution of each key component within the HPER-D3QN algorithm to overall performance, this study designed four ablation configurations—No-DTPA, No-HPER, No-Dueling, and No-Double—and compared them with the complete model (baseline) under identical simulation settings. The ablation experiments were conducted in the most challenging scenario, involving a high-density integrated combat airspace with 15 unmanned aircraft and 10 manned aircraft, under the highest uncertainty level (Level 5). As shown in Table 9, the removal of any

module led to performance degradation, with the absence of the HPER and DTPA modules having the most significant impact. Specifically, the success rate dropped by 9.27% and 8.98%, task completion time increased by 4.86% and 3.88%, and the FHP surged by 87.26% and 92.54%, respectively. These ablation experiments further validated the effectiveness of the proposed HPER-D3QN algorithm.

4.4. Generalization testing on high-fidelity simulation platform

To validate the generalization capability of the proposed algorithm, simulation experiments were conducted in typical scenarios using the model trained by the HPER-D3QN algorithm on a high-fidelity virtual simulation platform. This simulation platform, developed based on the Unity3D engine, constructs a high-fidelity battlefield airspace simulation system capable of accurately modeling the dynamic characteristics of both manned and unmanned aircraft, as well as various complex battlefield environmental factors, as shown in Fig. 15.

The simulation scenario focused on a joint operational setting, comprising 10 manned aircraft and 10 unmanned aircraft

operating simultaneously at identical flight altitudes according to predefined mission trajectories. Throughout these tests, UAVs equipped with the model trained by the HPER-D3QN algorithm autonomously navigated from their initial positions, dynamically avoiding potential collisions with both manned and unmanned aircraft from distinct combat units, ultimately reaching their designated mission objectives. A detailed depiction of the testing procedure is provided in Fig. 16.

Specifically, at $t = 125$ s, the UAV detected a potential conflict with an approaching manned aircraft (MA4) and immediately executed a right-turn maneuver to avoid the manned aircraft, as shown in Figs. 16(a) and 16(b). At $t = 216$ s, the UAV encountered another UAV (UAV5) on its left during flight and effectively avoided the collision by accelerating its right-turn maneuver, as depicted in Figs. 16(c) and 16(d). Finally, at $t = 295$ s, the UAV successfully reached the designated mission target, as illustrated in Figs. 16(e) and 16(f). The above experimental results demonstrated the strong generalization capability of the HPER-D3QN algorithm in complex joint operational environments. Furthermore, the outcomes emphasized the algorithm's effectiveness in ensuring successful mission execution, thus highlighting its significant potential and practical applicability in real-world battlefield scenarios.

5. Conclusions

To address the problem of adaptive collision avoidance for UAV in joint operational airspace, this paper proposes the HPER-D3QN algorithm based on the autonomous decision-making requirements of UAV during mission execution. By comprehensively analyzing the task scenarios where manned and unmanned aircraft operate together in a joint combat environment, the problem is formulated as an MDP. A partial observation model based on detection sectors and a DTPA are designed, leading to the construction of a well-structured observation space, action space, and reward function. Subsequently, a simulation environment is established, and the UAV's autonomous collision avoidance strategy is trained and evaluated. The experimental results demonstrate that:

- (1) The HPER-D3QN algorithm enables the trained agents to achieve adaptive collision avoidance and maneuver guidance for mission execution across battlefield environments with varying numbers, proportions, and uncertainty levels of manned and unmanned aircraft;
- (2) Ablation experiments confirm that the proposed DTPA algorithm and the HPER mechanism are critical factors in improving overall algorithm performance, with the HPER mechanism contributing most significantly;
- (3) Compared with other algorithms, HPER-D3QN exhibits clear advantages in terms of learning efficiency, adaptability, and robustness. Furthermore, the algorithm was successfully transferred and tested in a high-fidelity battlefield airspace simulation environment, further verifying its strong generalization capability and potential for practical application.

This study addresses a critical gap in both the theoretical and applied aspects of adaptive UAV collision avoidance within joint airspace. Nevertheless, the current validation is confined to simulation, which cannot fully capture the complexity and uncertainty of real-world scenarios. Future work will therefore focus on hardware-in-the-loop semi-physical simulations and flight tests under the Live-Virtual-Constructive framework [49], with an emphasis on the simulation-to-reality transfer mechanism and its engineering feasibility, laying the groundwork for practical deployment.

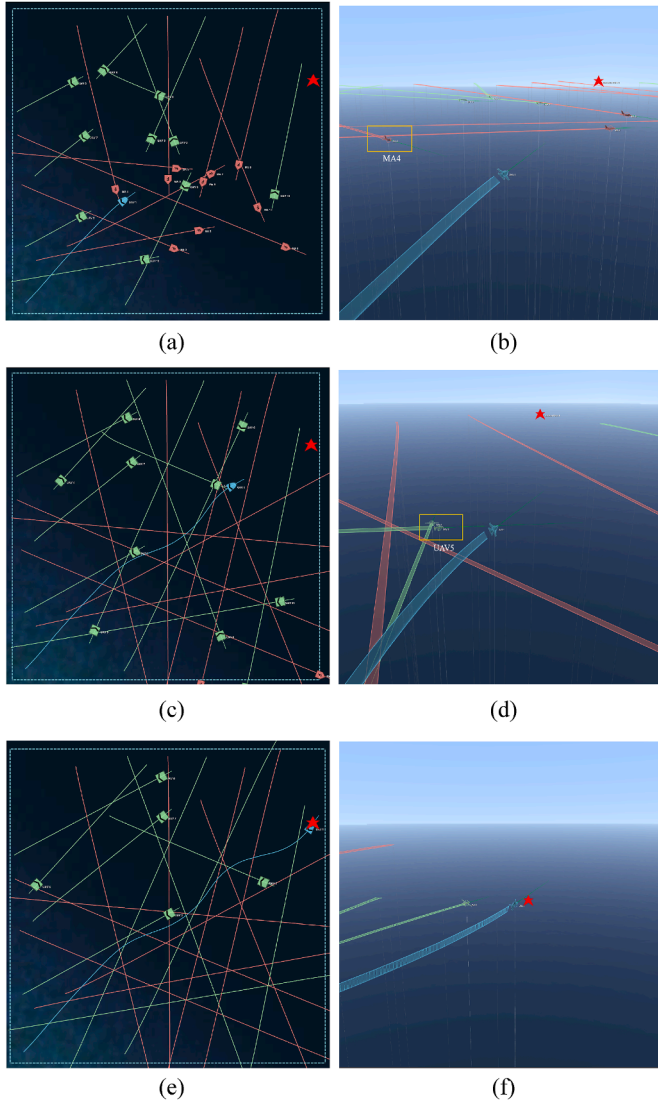


Fig. 16. Trajectories of UAV in joint operational airspace scenario. (a) $t = 125$ s (2D view); (b) $t = 125$ s (3D view); (c) $t = 216$ s (2D view); (d) $t = 216$ s (3D view); (e) $t = 295$ s (2D view); (f) $t = 295$ s (3D view).

CRediT authorship contribution statement

Yan Shen: Writing – original draft, Methodology. **Xuejun Zhang:** Writing – review & editing, Project administration, Conceptualization. **Yan Li:** Validation, Data curation. **Weidong Zhang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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